Name: ______________________ **Exit quiz**

Brackets in equations

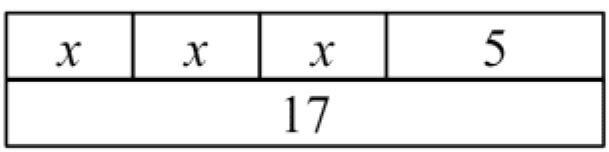
1 If you multiply both sides of an equation by 2, you will maintain ________________ .
Tick **1** correct answer

- ☐ the value
- ☐ equality
- ☐ equation
- ☐ the expressions

2 True or false? Multiplying by 5 and then adding 10 is the same as adding 10 then multiplying by 5. Tick **1** correct answer

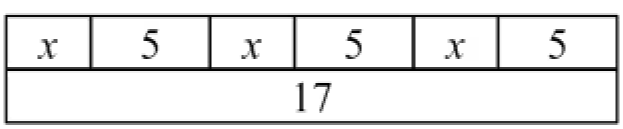
- ☐ True. Both have $\times 5$ and $+ 10$ so they are the same function.
- ☐ False. Order matters.

3 What equation does this bar model represent? Tick **2** correct answers



- ☐ $3 \times x + 5 = 17$
- ☐ $x + 5 \times 3 = 17$
- ☐ $(x + 5) \times 3 = 17$
- ☐ $x + 5 = 17$
- ☐ $3x + 5 = 17$

4 What does this bar model represent? Tick **3** correct answers



- ☐ $3 \times x + 5 = 17$
- ☐ $(x + 5) \times 3 = 17$
- ☐ $3(x + 5) = 17$
- ☐ $x + 5 \times 3 = 17$
- ☐ $3x + 15 = 17$

5 Multiply both sides of this equation $\frac{1}{3}x - 5 = 2x + 7$ by 3. Tick **2** correct answers

- [ ] $x - 5 = 6x + 7$
- [ ] $x - 15 = 6x + 21$
- [ ] $3x - 15 = 6x + 21$
- [ ] $3(\frac{1}{3}x - 5) = 3(2x + 7)$

6 Which of the below is the most efficient first step to solve the equation $\frac{3}{x} = 10$? Tick **1** correct answer

- [ ] $3(\frac{3}{x}) = 3(10)$
- [ ] $x(\frac{3}{x}) = x(10)$
- [ ] $10(\frac{3}{x}) = 10(10)$
- [ ] $10x(\frac{3}{x}) = 10x(10)$